

# GOOD EVENING EVERYONE

↳ We will start @ 09:10 pm

↪ By 09:23 PM

## First Index And Last Index



Success Rate: 92.86% Max Score: 10 Difficulty: Medium

Solve Challenge

## Print Alternate Array Elements Linewise

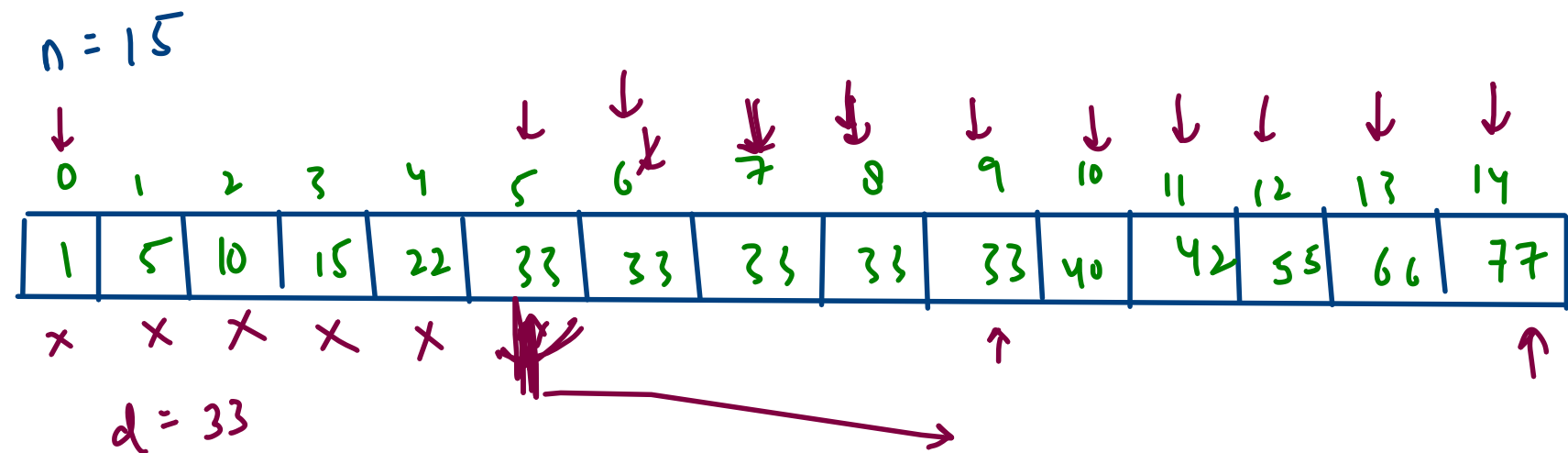


Success Rate: 100.00% Max Score: 10 Difficulty: Medium

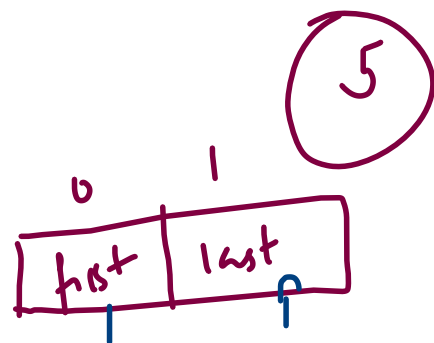
Solve Challenge

$n = 15$   
 15  
 1  
 5  
 10  
 15  
 22  
 33  
 33  
 33  
 33  
 40  
 42  
 55  
 66  
 77  
 33

all  
 first occurrence  
 last occurrence

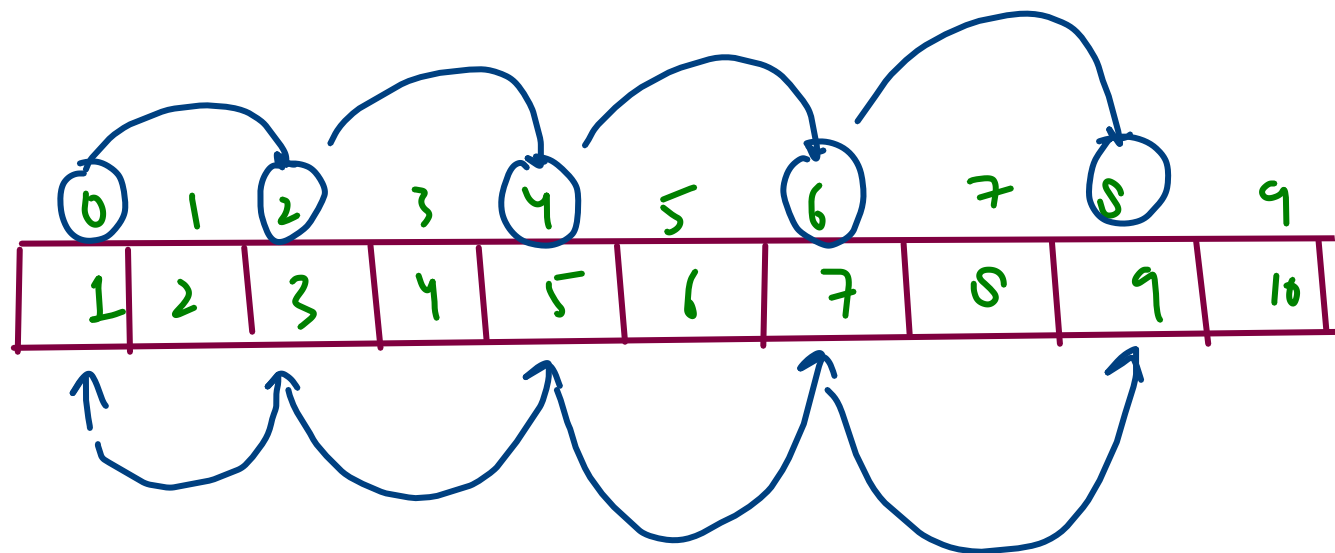


first  $\Rightarrow 5$   
 last = ~~5~~ ~~6~~ ~~7~~ ~~8~~ 9



$n = 10$

10  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10



1 3 5 7 9

$i = 1$      $i = 2$

for(int  $i = 0$ ;  $i < n$ ;  $i++$ )

0  
2  
4  
6  
8  
10

//

new  
↓  
m/m always  
allocated at  
next level

# Memory Management of Array

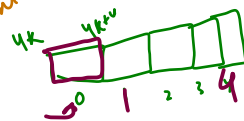
n=5.

int arr [ ] = new int [n];

int arr [ ];  
arr = new int [n];

index.  
[0, n-1]  
n ↗

1 int = 4 bytes.  
5 int = 4 x 5 = 20 bytes.



Index = how far a particular ele is  
from the base address

$$\Rightarrow \text{index} = \frac{\text{element address} - \text{base address}}{4}$$

address of array.  
base address.



Stack  
Memory.

arr[i] = 16  
↑  
arr[4K + 2x4] = 16  
arr[4K + 8] = 16  
i = 2

$$\Rightarrow \frac{4K - 4K}{4} \Rightarrow 0$$

$$\frac{4K + 4 - 4K}{4} \Rightarrow \frac{4}{4} = 1$$

??

$$\frac{4K + 4 - 4K}{4} = 1$$

arr[6] = 201

arr[4K + 24] → Arrays Out of Bounds.

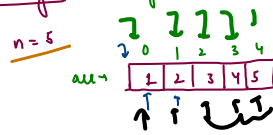
arr[4] = 200

arr[4K + 4x4] = 200

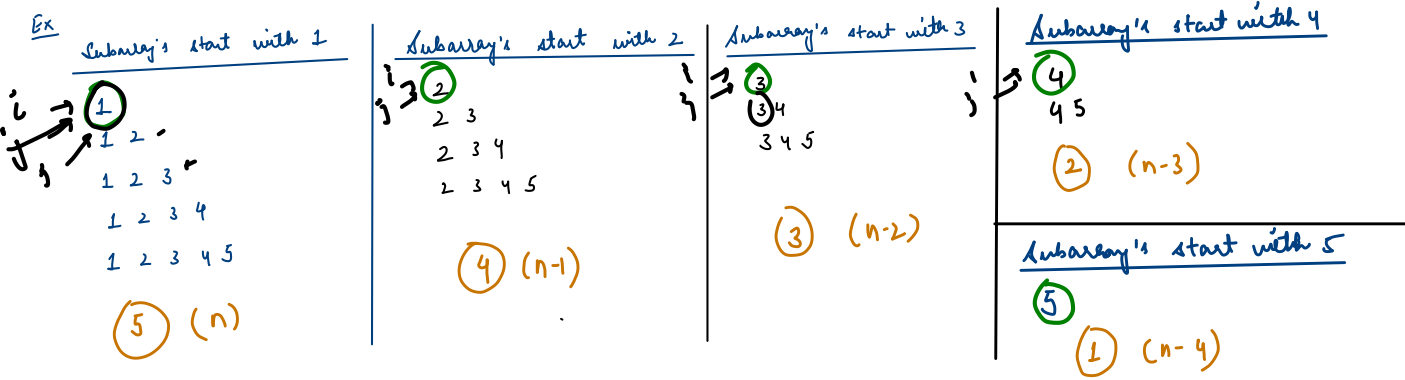
arr[4K + 16] = 200.

Subarray  $\rightarrow$  any continuous part of an array.

$n \rightarrow$



Ex



$5+4+3+2+1 \Rightarrow (15)$

$\Rightarrow n + (n-1) + (n-2) + (n-3) + (n-4) + \dots + 1$

Sum of 'n' natural nos.

$\hookrightarrow \frac{n(n+1)}{2}$  where n is size of an array.

1<sup>st</sup> loop  $\rightarrow$  Iterating over the array considering as a starting pt.

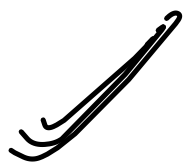
2<sup>nd</sup> loop  $\rightarrow$  Stopping pt. of every subarray

3<sup>rd</sup> loop  $\rightarrow$  Printing the subarray ( $i=j$ ),

```

public static void printSubArrays(int arr[]) {
    int n = arr.length;
    for(int i=0; i<n; i++) { →
        for(int j=i; j<n; j++) {
            for(int k = i; k<=j; k++) { →
                System.out.print(arr[k] + " ");
            }
            System.out.println();
        }
    }
}

```



1  
 1 2  
 1 2 3  
 1 2 3 4  
 1 2 3 4 5  
 2  
 2 3

2 3 4  
 2 3 4 5  
 3  
 3 4  
 3 4 5  
 4  
 4 5

5

n = 5

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 0 | 1 | 2 | 3 | 4 |

$i=0 < 5$   
 $j=0 < 5$   
 $k=0 <= 0$   
 $k=1 <= 0 (F)$   
 $j=1 < 5$   
 $k=0 <= 1$   
 $k=1 <= 1$   
 $k=2 <= 1 (F)$

$j=2 < 5$   
 $k=0 <= 2$   
 $k=1 <= 2$   
 $k=2 <= 2$   
 $k=3 <= 2$

$j=3 < 5$   
 $k=0 <= 3$   
 $k=1 <= 3$   
 $k=2 <= 3$   
 $k=3 <= 3$   
 $k=4 <= 3$

$j=4 < 5$   
 $k=0 <= 4$   
 $k=1 <= 4$   
 $k=2 <= 4$   
 $k=3 <= 4$   
 $k=4 <= 4$   
 $k=5 <= 4$

$j=5 < 5 (F)$   
 $i=1 < 5 (T)$   
 $j=1 < 5 (i)$   
 $k=0 <= 1$   
 $k=2 <= 1 (F)$   
 $j=2 < 5$   
 $k=0 <= 2$   
 $k=2 <= 2$   
 $k=3 <= 2$

$j=3 < 5$   
 $k=0 <= 3$   
 $k=2 <= 3$   
 $k=3 <= 3$   
 $k=4 <= 3 (F)$   
 $j=4 < 5$   
 $k=1 <= 4$   
 $k=2 <= 4$   
 $k=3 <= 4$   
 $k=4 <= 4$   
 $k=5 <= 4$   
 $j=5 < 5$   
 $i=2 < 5$